

Testing Market Efficiency using Statistical Arbitrage with Applications to Momentum and Value Strategies

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Abstract

This paper introduces the concept of statistical arbitrage, a long horizon trading opportunity that generates a riskless profit and is designed to exploit persistent anomalies. Statistical arbitrage circumvents the *joint hypothesis* dilemma of traditional market efficiency tests because its definition is independent of any equilibrium model and its existence is incompatible with market efficiency. We provide a methodology to test for statistical arbitrage and then empirically investigate whether momentum and value trading strategies constitute statistical arbitrage opportunities. Despite controlling for transaction costs and the influence of small stocks, we find evidence that these strategies generate statistical arbitrage. Furthermore, their profitability does not appear to decline over time.

1 Introduction

A large number of empirical studies conclude that stock prices appear to contradict the efficient markets hypothesis. For example, Jegadeesh and Titman (1993) investigate a trading strategy

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that buys well-performing stocks and sells poor-performing stocks. They document average excess returns of 12% per year, where excess returns are defined relative to a standard capital asset pricing model. Lakonishok, Shleifer, and Vishny (1994) reach a similar conclusion via buying *value* and selling *glamour* stocks identified with variables such as price earnings ratios, dividends, book to market values, cash flows, and sales growth. Chan, Jegadeesh, and Lakonishok (1996) confirm the excess returns of portfolios formed on the basis of past returns and earnings announcements. Profitable trading strategies based on exchange rate movements for US firms with overseas operations are found in Bartov and Bodnar (1994). Michaely, Thaler, and Womack (1995) document profitable trading strategies involving dividend announcements and omissions. Ikenberry, Lakonishok, and Vermaelen (1995) report abnormal returns following open market share repurchases, while Loughran and Ritter (1995) document the long term underperformance of firms issuing equity.¹

However, all the above-mentioned studies suffer from the joint hypothesis problem. Fama (1998) cautions against rejecting market efficiency because previous tests are contingent upon a specific model for equilibrium returns. Therefore, abnormal returns need not imply the rejection of market efficiency. Instead, they may indicate that the equilibrium model is misspecified. According to Fama (1998) most long-term anomalies are also sensitive to the statistical methodology utilized and therefore inferences drawn using long-term returns (e.g. over 5 years) are circumspect. This paper designs a statistical methodology that addresses Fama's criticisms and circumvents the joint hypothesis problem.

The methodology proposed in this paper is based on the concept of a statistical arbitrage. Roughly speaking, a statistical arbitrage is a long horizon trading opportunity that generates a riskless profit. As such, statistical arbitrage is a natural extension of the trading strategies utilized in the existing empirical literature on anomalies. Statistical arbitrage is the time series analogue of the limiting arbitrage opportunity contained in Ross (1976). More importantly, statistical arbitrage is defined without reference to any equilibrium model, therefore, its existence is inconsistent with market equilibrium and, by inference, market efficiency (see Jarrow [1988], chapter 19). As such, the notion of statistical arbitrage enables the rejection of market efficiency without invoking the

¹Shleifer (2000) provides an excellent comprehensive review of these studies.

joint hypothesis of an equilibrium model. Instead, the joint hypothesis of an equilibrium model is replaced with an assumed stochastic process for trading profits. This paper provides a statistical test to determine whether trading profits generated by a persistent anomaly constitute statistical arbitrage based on *incremental trading profits* computed over short time horizons.

We apply this new methodology to test whether momentum and value trading strategies are consistent with statistical arbitrage. These trading strategies are tested under the assumption that the expected trading profits are constant over time. We refer to this as the constrained mean (CM) test for statistical arbitrage. We test two implications of statistical arbitrage opportunities: one, whether the trading strategy's expected profits are greater than zero; and two, whether the strategy's time-averaged variance decreases over time. We also compute the probability of a loss for these trading strategies, providing additional insights into whether these strategies represent statistical arbitrage.

To minimize data mining concerns, the momentum strategies we investigate are modeled after the momentum strategies tested in Jegadeesh and Titman (1993) while the value strategies are modeled after the contrarian strategies of Lakonishok, Shleifer, and Vishny (1994). Specifically, we examine an exhaustive set of 16 momentum strategies. As in Jegadeesh and Titman (1993), these strategies have formation periods and holding periods of 3, 6, 9, and 12 months. In addition, we also examine an exhaustive set of 12 value strategies. As in Lakonishok, Shleifer, and Vishny (1994), value strategies are based on book-to-market equity, cashflow-to-price, and earnings-to-price ratios as well as three-year sales growth with one, three and five year holding periods. The sample period is from January 1965 to December 2000 and subsumes that of both Jegadeesh and Titman (1993) and Lakonishok, Shleifer, and Vishny (1994).

Our test of statistical arbitrage reveals that of the 16 momentum trading strategies, six produce statistical arbitrage at the 5% level with three more at the 10% level. The six momentum strategies that produce statistical arbitrage at the 5% level include two strategies with a formation period of three months, three strategies with a formation period of six months and one strategy with a formation period of nine months. The momentum strategy that longs the highest return decile and shorts the lowest return decile based on six months of past returns and holds that portfolio for

twelve months most strongly rejects the null hypothesis of no statistical arbitrage. The probability of making a loss on this strategy falls below 1% after just 86 months of trading. Of the 12 value strategies we examine, five strategies yield statistical arbitrages at the 5% level. The convergence rates for value strategies to riskless arbitrages are comparable to those of momentum strategies. After just 79 months of trading, the probability of incurring a loss on the value strategy using the past three years of sales growth with a holding period of one year falls below 1%. Interestingly, while we find that the expected profit of the trading strategies based on book-to-market equity (with holding periods of one year and three years) are significantly positive, their time-averaged variances do not decline. Hence, they do not constitute statistical arbitrage opportunities (at the 5% level). In summary, momentum strategies and value strategies provide strong evidence against the efficient markets hypothesis.

Our first robustness check evaluates whether the assumption of constant expected profits is appropriate. We compare the constrained mean (CM) model with an unconstrained mean (UM) model that incorporates an additional parameter allowing the expected profits to vary over time. Three independent tests confirm that the CM model is appropriate. Firstly, the rates of change for expected profits are usually not statistically different from zero. Of the 15 momentum and value strategies which are CM statistical arbitrages at the 10% or 5% level, only one registers a statistically negative rate of decay in its expected profits. The profits from momentum and value strategies do not appear to erode over time. Secondly, the insample fit as measured by the root mean squared error (RMSE) for the CM and UM models are indistinguishable, indicating there is no advantage to using the more complicated UM model. Thirdly, the sum of squared normalized residuals are almost identical. Hence, the UM model spreads the information contained in trading profits over an additional variable without offering an improved fit, thereby weakening the power of the statistical arbitrage test.

Another possible concern is that market inefficiencies may disappear after including transactions costs. To investigate this issue, we compute the turnover of each portfolio and combine these results with the estimated round trip transactions costs of Chan and Lakonishok (1997). We adjust the incremental profits downward by these transactions costs. Even after this adjustment, all of the

11 statistical arbitrages at the 5% level remain statistical arbitrages at the 5% or 10% level. This suggests that the statistical arbitrages represent feasible and attractive trading opportunities for market participants.

A common feature of stock market anomaly studies is that small stocks are often less efficient than large stocks (Hong, Lim, and Stein, 2000; Mitchell and Stafford, 2000). Moreover since small stocks require greater transactions costs and are less liquid, we investigate whether our results survive their exclusion. Even after removing stocks with market equity below the 40% NYSE percentile from our sample, all the momentum portfolios and 10 out of 12 value portfolios still test positively for statistical arbitrage at the 5% level. Our previous results do not appear to be driven by small stocks.

Finally, to further ascertain the robustness of our statistical methodology, we perform a series of simulations incorporating possible misspecifications such as autocorrelation, jumps, and non-stationarity. Our statistical arbitrage test is then applied to these simulated processes. The simulations indicate that our statistical test is biased towards accepting the null hypothesis of no statistical arbitrage, and thus the acceptance of market efficiency. Consequently, instances of statistical arbitrage found herein are hard to reconcile with market efficiency.

This paper is organized as follows. Section 2 introduces the concept of statistical arbitrage and its relation to market efficiency. To illustrate the methodology, statistical tests for the existence of statistical arbitrage are derived in Section 3. The data and methodology used in our empirical study is given in Section 4 while Section 5 discusses our empirical results. The robustness of our results is the topic of Section 6 while the conclusion follows in Section 7.

2 Statistical arbitrage and market efficiency

This section introduces the notion of a statistical arbitrage and its relation to market efficiency. Let $(\Omega, F, (F_t : t \geq 0), P)$ be a filtered probability space over an infinite horizon $[0, \infty)$, satisfying the usual conditions. Here P is the statistical probability measure. Traded in the economy are a stock² S_t and money market account B_t initialized at a dollar ($B_0 = 1$). Any spot rate process

²As will become clear, the stock can be interpreted as a portfolio of traded assets.

consistent with an arbitrage free evolution is acceptable (see Heath, Jarrow, Morton (1992)).

Trading strategies are central to the notion of statistical arbitrage. Let the stochastic processes $(x(t), y(t) : t \geq 0)$ represent a *zero initial cost, self-financing trading* strategy involving $x(t)$ units of a stock and $y(t)$ units of a money market account at time t . This trading strategy is formulated using only available information. For example, trading strategies may involve rules based on past returns, firm size, earnings announcements, market versus book values, sales growth, or macroeconomic conditions. These trading strategies, by definition, must have zero initial cost ($x(0)S_0 + y(0) = 0$) and have no cash inflow or outflow after their construction (the self-financing condition). The stock could itself be a (zero cost) self-financing portfolio consisting of long and short positions in the risky assets.

Although models of market equilibrium may assist in identifying possible trading strategies, a model of market equilibrium is not required. Let the process $V(t)$ denote the cumulative trading profits at time t that are generated by such a trading strategy $(x(t) : t \geq 0)$.³ For some of the following analysis, it will be necessary to work with the discounted value of the cumulative trading profits defined as $v(t) = V(t)/B_t$. To illustrate these concepts and various characteristics of typical trading strategies, we consider a few examples.

Example 1 *Consider the standard Black-Scholes economy with a non-dividend paying stock price S_t that evolves according to*

$$S_t = S_0 e^{\alpha t - \sigma^2 t/2 + \sigma W_t}$$

and with a money market account satisfying

$$B_t = e^{rt}$$

where α, σ^2, r are non-negative constants, $\alpha > r$, and W_t is a standard Brownian motion.

Consider the zero initial investment, self-financing trading strategy that consists of buying and holding (forever) 1 unit of the stock, and financing it with the money market account. The value

³The process $V(t)$ is dollar denominated and it is neither a cumulative excess return nor a cumulative residual with respect to an equilibrium model.

of this portfolio at time t is

$$V(t) = 1 \cdot S_t - S_0 \cdot e^{rt} = S_0 \left(e^{[\alpha - \sigma^2/2]t + \sigma W_t} - e^{rt} \right)$$

representing the cumulative trading profits with an initial value $V_0 = 0$. The expectation and variance⁴ of the cumulative discounted trading profits both approach infinity as $t \rightarrow \infty$,

$$E^P[v(t)] = S_0 \left[e^{(\alpha - r)t} - 1 \right] \rightarrow \infty \quad \text{and} \quad \text{Var}^P[v(t)] = S_0^2 e^{2(\alpha - r)t} \left[e^{\sigma^2 t} - 1 \right] \rightarrow \infty$$

where $E^P[\cdot]$ is the expectation with respect to P and $\text{Var}^P[\cdot] = E[\cdot]^2 - (E[\cdot])^2$. Although expected profits increase without bound, there is an offsetting increase in their variance. In addition, the time-averaged variance $\frac{\text{Var}^P[v(t)]}{t}$ also converges to ∞ as $t \rightarrow \infty$, the significance of which will become apparent when statistical arbitrage is defined.

Many instances of premiums or “excess returns” in the anomalies literature are generated by portfolios that are long and short portfolios of stocks formed on the basis on certain cross sectional characteristics. The next example considers a portfolio of stocks whose dynamics involve arithmetic rather than geometric Brownian motion. Arithmetic Brownian motion is a reasonable hypothesis in this circumstance since a portfolio of long and short positions is not a limited liability asset.

Example 2 Consider the discounted cumulative value of a trading strategy $v(t)$ that evolves according to

$$v(t) = \alpha t + \sigma W_t$$

with $v(0) = 0$ (zero initial value). To exhibit negative values, the stock in this example must be interpreted as a portfolio of stocks, consisting of both long and short positions. The expectation and variance of the cumulative discounted trading profits are both unbounded as $t \rightarrow \infty$

$$E^P[v(t)] = \alpha t \rightarrow \infty \quad \text{and} \quad \text{Var}^P[v(t)] = \sigma^2 t \rightarrow \infty$$

while the time-averaged variance is finite and equals σ^2 .

⁴See Casella and Berger (1990) page 628 for the variance of a lognormal random variable.

The previous examples illustrate typical trading strategies that do not represent long horizon “excess profit” opportunities. This is because as the trading strategies’ expected profits increase across time, there is a corresponding increase in the trading profit’s variance, and even when time-averaged the variance does not disappear. In contrast, the following example captures the intuitive notion of a statistical arbitrage.

Example 3 *Let us revisit the buy and hold trading strategy underlying Example 2. Suppose that the discounted trading profits over an arbitrary intermediate trading interval $[t_{k-1}, t_k]$ can be represented as*

$$v(t_k) - v(t_{k-1}) = \mu + \sigma z_k$$

where $\mu, \sigma > 0$ and z_k are i.i.d. random variables with zero mean and variance $1/k$. This trading strategy has positive expected discounted profits over every interval (μ), but with random noise (σz_k) appended. The variance of the noise is decreasing over time. Once again, $v(t_0) = 0$ and the cumulative discounted trading profits at time t_n equals

$$v(t_n) = \sum_{k=1}^n [v(t_k) - v(t_{k-1})] = \mu n + \sigma \sum_{k=1}^n z_k.$$

Note that $E^P[v(t_n)] = \mu n$ and $Var^P[v(t_n)] = \sigma^2 \sum_{k=1}^n \frac{1}{k}$. Therefore, $\frac{Var^P[v(t_n)]}{n} = \sigma^2 \frac{\sum_{k=1}^n \frac{1}{k}}{n} \rightarrow 0$ as $n \rightarrow \infty$. It can also be shown⁵ that $P(v(t_n) < 0) \rightarrow 0$, implying that the discounted trading profits are non-negative in the limit.

Across time, analogous to cross-sectional diversification in Ross’ APT, the random noise in this trading strategy is “diversifiable,” thereby generating a limiting arbitrage opportunity. Indeed, this trading strategy has a positive discounted expected profit with a time-averaged variance approaching zero. Over a long time horizon, this trading strategy captures the notion of a “statistical arbitrage.”

Given these insights, we now formalize the notion of a statistical arbitrage.

⁵These dynamics are a special case of Assumption 1 with the result proved in Appendix A.

Definition 4 A statistical arbitrage is a zero initial cost, self-financing trading strategy $(x(t) : t \geq 0)$ with cumulative discounted value $v(t)$ such that:

1. $v(0) = 0$
2. $\lim_{t \rightarrow \infty} E^P[v(t)] > 0$
3. $\lim_{t \rightarrow \infty} P(v(t) < 0) = 0$, and
4. $\lim_{t \rightarrow \infty} \frac{Var^P[v(t)]}{t} = 0$ if $P(v(t) < 0) > 0 \quad \forall t < \infty$.

By definition, a statistical arbitrage satisfies four conditions (i) it is a zero initial cost ($v(0) = 0$) self-financing trading strategy, that in the limit has (ii) positive expected discounted profits, (iii) a probability of a loss converging to zero, and (iv) a *time-averaged* variance converging to zero if the probability of a loss does not become zero in finite time. That is, the fourth condition only applies when there always exists a positive probability of losing money.⁶ Otherwise, if $P(v(T) < 0) = 0$ for some time $T < \infty$, then a standard arbitrage opportunity is available. In economic terms, the fourth condition implies that a statistical arbitrage opportunity eventually produces riskless incremental profit, with an associated “Sharpe” ratio increasing monotonically through time. This is consistent with the variance of the trading strategy increasing towards infinity with time, but with a “growth rate” less than linear. Condition 4 also implies that $P(v(t)/\sqrt{t} < 0) \rightarrow 0$, i.e. the risk of losing money (per unit time) becomes zero in the limit.⁷ As indicated, the ability of a trading strategy to reduce its average variance over time by adjusting either the composition⁸ or

⁶An additional justification for the fourth condition is given in Appendix B.

⁷Note that $Var^P[v(t)/\sqrt{t}] \rightarrow 0$, which implies that $v(t)/\sqrt{t} - E^P[v(t)/\sqrt{t}] \rightarrow 0$ in probability, i.e. $P(|v(t)/\sqrt{t} - E^P[v(t)/\sqrt{t}]| \geq \epsilon) \rightarrow 0$ for all $\epsilon > 0$. Hence, $P(-v(t)/\sqrt{t} + E^P[v(t)/\sqrt{t}] \geq \epsilon) \rightarrow 0$ for all $\epsilon > 0$. But, by condition 2, for large enough t , $E^P[v(t)/\sqrt{t}] \geq 0$. Thus, for all $\epsilon > 0$, $P(v(t)/\sqrt{t} \leq -\epsilon) < P(v(t)/\sqrt{t} - E^P[v(t)/\sqrt{t}] \leq -\epsilon) \rightarrow 0$ and this proves the claim.

⁸There does not exist a unique transformation from non self-financing trading strategies in the anomalies literature into self-financing trading strategies. Furthermore, the self-financing condition may change the composition of the portfolio over time. For example, profits may be invested in the riskfree money market account, creating a potential interaction between the self-financing and the decreasing time-averaged variance properties.

magnitude of its long and short positions is essential to generate statistical arbitrage.⁹

A standard arbitrage opportunity is a special case of statistical arbitrage. Indeed, a standard arbitrage has $V(0) = 0$ where there exists a finite time¹⁰ $T > 0$ such that $V(T)$ satisfies $P(V(T) > 0) > 0$ and $P(V(T) \geq 0) = 1$. To transform the standard arbitrage opportunity into an infinite horizon self-financing trading strategy, we just invest the proceeds at time T into the money market account, i.e. $V(s) = V(T) \frac{B_s}{B_T}$ for $s \geq T$. Note that $v(s) = V(T) \frac{B_s}{B_T} \frac{1}{B_s} = v(T)$. Then, $\lim_{s \rightarrow \infty} E^P[v(s)] = E^P[v(T)] > 0$ which satisfies condition 2 and $\lim_{s \rightarrow \infty} P(v(s) < 0) = P(v(T) < 0) = 0$ which satisfies condition 3 and implies that condition 4 is not applicable. However, this paper's intent is to construct a statistical test for evaluating market efficiency with regards to *persistent anomalies*, trading strategies whose probability of losing money is always positive.

This definition is similar to the limiting arbitrage opportunity used to construct Ross' APT (1976). The difference between the two concepts is that a statistical arbitrage is a limiting condition across time, while Ross' APT is a cross-sectional limit at a point in time. This difference necessitates discounting by the money market account and the normalization by time in condition 4. Therefore, just as Ross' APT is appropriate in an economy with a "large" number of assets, our methodology is appropriate for "long" time horizons. It is important to emphasize that investors need not wait until infinity to benefit from a statistical arbitrage. Investors with long but finite time horizons view statistical arbitrage opportunities as "too good to pass up" as they offer positive expected discounted profit, variance (per unit time) that becomes arbitrarily small, and decreasing risk of a loss. Indeed, although statistical arbitrage is defined over an infinite time horizon, there exists a finite timepoint t^* such that the probability of a loss is arbitrarily small, $P(v(t^*) < 0) = \epsilon$. Thus, comparing an arbitrage opportunity at time t^* having the property $P(v(t^*) < 0) = 0$ with a statistical arbitrage opportunity reveals they are only separated by an ϵ probability of a loss.

It is well known that arbitrage opportunities are incompatible with an efficient market, see Jensen (1978) for a general discussion¹¹. Tests of market efficiency in the options market based

⁹The fourth property of statistical arbitrage also partially mitigates the concerns of Shleifer and Vishny (1997) with respect to "risky" arbitrage.

¹⁰Note that $T = 0_+$ is possible, with $V(0_+) = \alpha > 0$ for $\alpha \in R_{++}$. This would correspond to a mispricing locked in at time 0 with no future cash flows.

¹¹Jensen states that the existence of economic trading profits (defined therein on page 96) reject an efficient market.

on the absence of arbitrage are numerous. For example, Klemkosky and Resnick (1979) and Kamara and Miller (1995) study violations of put-call parity, while Bodurtha and Courtadon (1986) investigate the efficiency of foreign currency options. No arbitrage in the box spreads of options is conducted by Ronn and Ronn (1989), and after the 1987 crash by Hemler and Miller (1997). Tests for statistical arbitrage rejecting market efficiency can be viewed as an extension of this literature. Indeed, statistical arbitrage rejects the market as being in *any economic equilibrium*, an important prerequisite for an efficient market (see Jarrow (1988), chapter 19). A statistical arbitrage (analogous to an arbitrage opportunity in Ross' APT) will induce investors to engage in trading, and consequently prevent the market from reaching an equilibrium. Even a single trader operating under a finite time horizon and willing to pursue a statistical arbitrage is sufficient to conclude these trading opportunities reject market efficiency.

Our idea of extending standard arbitrage to its infinite horizon counterpart is related to a literature extending simple arbitrage in alternative ways. In the context of incomplete markets, Cochrane and Saa-Requejo (2000) invoke Sharpe ratios to find asset prices that are *good deals* to investors while Bernardo and Ledoit (2000) exclude investments whose maximum gain-loss ratios are too *attractive*. Both of these approaches also investigate trading opportunities that generalize the definition of arbitrage without specifying a particular model of market equilibrium. Carr, Geman, and Madan (2001) introduce the concept of *acceptable* opportunities which would be executed by any reasonable investor, implying their existence is incompatible with an efficient market. Statistical arbitrage is a similar generalization of arbitrage, but differs from the previous work as it is defined under the observed statistical measure rather than a collection of probability measures.

This notion of statistical arbitrage can also be extended to study the traditional weak- and strong-form tests of market efficiency. Indeed, in constructing the trading strategies investigated, weak-form statistical arbitrage would utilize only publicly available information, while strong-form statistical arbitrage would utilize all available information.

An arbitrage opportunity (and a statistical arbitrage) satisfies his definition of economic trading profits.

3 Tests for statistical arbitrage

To test for statistical arbitrage, a time series of dollar denominated discounted cumulative trading profits $v(t_1), v(t_2), \dots, v(t_n)$ generated by a trading strategy are analyzed. For a given trading strategy, let $\Delta v_i = v(t_i) - v(t_{i-1})$ denote increments of the discounted cumulative trading profit measured at equidistant time points $t_i - t_{i-1} = \Delta$ with $t_i = i\Delta$.

Our test for statistical arbitrage requires an assumed process for the underlying asset prices in the economy. Instead of imposing the stochastic process directly on the traded asset prices as in the above example, it is more convenient to impose an assumption¹² on the stochastic process describing discounted incremental trading profits Δv_i . A general class of processes for the incremental trading profit is described by a discrete time semi-martingale $\Delta v_i = \mu(i)\Delta + \sigma(i)Z_i$ with a martingale component Z_i and coefficients $\mu(i)$, $\sigma(i)$. For simplicity of the initial presentation, we impose the assumption that the trading profits have independent incremental trading profits. This assumption is later relaxed but empirical results confirm that it does not alter our conclusions regarding market efficiency. For greater generality, we also allow the expected profits of the trading strategy to vary over time. Thus, we begin our analysis with the unconstrained mean (UM) model. The quantity Δ denoting the time between equidistant increments may be set to one without any loss of generality with t_i understood as being i .

Assumption 1 *Let the discounted incremental trading profits¹³ satisfy*

$$\Delta v_i = \mu i^\theta + \sigma i^\lambda z_i \tag{1}$$

for $i = 1, 2, \dots, n$ where z_i are i.i.d $\mathcal{N}(0, 1)$ random variables with $z_0 = 0$. Note that both $v(t_0)$ and Δv_0 are zero.

It is easy to see that the expectation and variance of the discounted incremental trading profits are $E[\Delta v_i] = \mu i^\theta$ and $Var[\Delta v_i] = \sigma^2 i^{2\lambda}$. Observe that for $\lambda < 0$, the variance of Δv_i decreases

¹²In theory, discounted incremental trading profits derived from portfolios are well represented by a normal distribution since the impact of idiosyncratic jumps are mitigated.

¹³Note that Assumption 1 implies that condition 4 of the definition of statistical arbitrage is relevant. It excludes standard arbitrage opportunities, whose existence can be tested via alternative and well-known procedures.

over time, which ultimately satisfies the 4th condition of statistical arbitrage as seen later. Also note that the dynamics of the discounted incremental trading profit in equation (1) with $\lambda < 0$, $\mu > 0$, and $\theta \geq 0$ does not advocate a trading strategy based on waiting for the volatility to decline (and perhaps the mean to increase) before investing. Instead, it is optimal for investors faced with such an opportunity to immediately begin trading and earn a positive expected profit while enjoying the benefit of decreasing time-averaged variance.

The use of normal increments in Assumption 1 is justified by the central limit theorem (CLT). The discounted cumulative trading profit $v(t)$ is, by definition, the sum of the Δv_i 's and a normalized sum of increments results in an asymptotically normal random variable, often with rapid convergence, under mild regularity¹⁴ conditions. For example, assume the same coefficients for the increments,

$$\Delta v_i = \mu i^\theta + \sigma i^\lambda Z_i$$

but suppose Z_i are not normally distributed. Aggregating over observations forms a sequence of *average* incremental trading profits $\overline{\Delta v_j}$ for $j = 1, 2, \dots, m < n$ that better conform to the assumption of normality. The same estimation and testing methodologies presented for the normal case remain applicable for the averaged observations. Thus, for large n , the assumption of normality is relatively innocuous.

The discounted cumulative trading profits generated by the trading strategy are

$$v(t_n) = \sum_{i=1}^n \Delta v_i \stackrel{d}{\sim} \mathcal{N}\left(\mu \sum_{i=1}^n i^\theta, \sigma^2 \sum_{i=1}^n i^{2\lambda}\right) \quad (2)$$

while the log likelihood function for the increments in equation (1) equals

$$\log L(\mu, \sigma^2, \lambda, \theta | \Delta v) = -\frac{1}{2} \sum_{i=1}^n \log(\sigma^2 i^{2\lambda}) - \frac{1}{2\sigma^2} \sum_{i=1}^n \frac{1}{i^{2\lambda}} (\Delta v_i - \mu i^\theta)^2 \quad (3)$$

allowing maximum likelihood estimation to generate the four required parameters. The score equations required to solve for the four parameters are found¹⁵ in the next lemma.

¹⁴Resnick (1998) chapter 9 contains relevant material including the Lindeberg-Feller CLT on page 315. Regularity conditions that satisfy the *Lindeberg condition* include uniform asymptotic negligibility or finite second moment.

¹⁵The property $\frac{\partial}{\partial x} t^{\alpha x} = \alpha \log(t) t^{\alpha x}$ is used repeatedly when solving for $\hat{\lambda}$ and $\hat{\theta}$.

Lemma 5 *Parameter estimates are obtained by solving the following four equations for the four unknown parameters.*

$$\frac{\partial \log L(\mu, \sigma^2, \lambda, \theta | \Delta v)}{\partial \mu} : \hat{\mu} = \frac{\sum_{i=1}^n \Delta v_i i^{\hat{\theta}-2\hat{\lambda}}}{\sum_{i=1}^n i^{2(\hat{\theta}-\hat{\lambda})}} \quad (4)$$

$$\frac{\partial \log L(\mu, \sigma^2, \lambda, \theta | \Delta v)}{\partial \sigma^2} : \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \frac{1}{i^{2\hat{\lambda}}} \left(\Delta v_i - \hat{\mu} i^{\hat{\theta}} \right)^2 \quad (5)$$

$$\frac{\partial \log L(\mu, \sigma^2, \lambda, \theta | \Delta v)}{\partial \theta} : \sum_{i=1}^n \Delta v_i \log(i) i^{\hat{\theta}-2\hat{\lambda}} = \hat{\mu} \sum_{i=1}^n \log(i) i^{2(\hat{\theta}-\hat{\lambda})} \quad (6)$$

$$\frac{\partial \log L(\mu, \sigma^2, \lambda, \theta | \Delta v)}{\partial \lambda} : \hat{\sigma}^2 \sum_{i=1}^n \log(i) = \sum_{i=1}^n \frac{\log(i)}{i^{2\hat{\lambda}}} \left(\Delta v_i - \hat{\mu} i^{\hat{\theta}} \right)^2 \quad (7)$$

As expected, for $\theta = 0$ and $\lambda = 0$, equations (4) and (5) in the previous lemma reduce to the standard normal MLE estimators

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n \Delta v_i \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\Delta v_i - \hat{\mu})^2.$$

Lemma 5 is easily reproduced for different mean and variance functions describing incremental trading profits. However, the functional form of the coefficients in equation (1) with gradual changes in the mean and variance offers an excellent description of realized trading profit increments in subsequent empirical testing. Moreover, misspecifying the incremental trading profit process only serves to increase the likelihood of accepting the null hypothesis of no statistical arbitrage. Therefore, evidence which indicates the existence of statistical arbitrage is robust to criticisms regarding the specified stochastic process. Standard errors for the parameters in Lemma 5 are obtained from the Hessian¹⁶ matrix evaluated at the estimated parameters to facilitate significance testing.

Under the previous assumption, the following theorem provides a test¹⁷ for the existence of

¹⁶The Hessian matrix is available from the authors by request.

¹⁷The issue of in-sample versus out-of-sample testing inherent in all statistical tests remains as it may be possible to devise a complicated trading strategy that only produces statistical arbitrage in a specific time period. Issues of stationarity are addressed in later simulations.

statistical arbitrage.

Theorem 6 *A trading strategy generates a statistical arbitrage with $1 - \alpha$ percent confidence if the following conditions are satisfied:*

$$H1: \mu > 0,$$

$$H2: \lambda < 0,$$

$$H3: \theta > \max \left\{ \lambda - \frac{1}{2}, -1 \right\},$$

with the sum of the p values for the individual tests forming an upper bound for the type I error α .

Proof. In order to satisfy the fourth condition of statistical arbitrage, equation (2) implies the parameter λ must be negative to ensure $\frac{\sigma^2 \sum_{i=1}^n i^{2\lambda}}{n} \rightarrow 0$ while the second condition requires the parameter μ to be positive. Any value of θ ensures $\mu \sum_{i=1}^n i^\theta > 0$ provided $\mu > 0$. However, the convergence of $P(v(t) < 0)$ to zero requires

$$\frac{\sum_{i=1}^n i^\theta}{\sqrt{\sum_{i=1}^n i^{2\lambda}}} \rightarrow \infty$$

with details in Appendix A. The leading order term of the sum in the numerator (using left endpoints)

$$\sum_{i=1}^n i^\theta \geq \int_1^n s^\theta ds = \frac{n^{1+\theta}}{1+\theta} - \frac{1^{1+\theta}}{1+\theta}$$

divided by the square root of the sum in the denominator (using right endpoints)

$$\sum_{i=1}^n i^{2\lambda} \leq \int_1^n (s+1)^{2\lambda} ds = \frac{(n+1)^{2\lambda+1}}{2\lambda+1} - \frac{2^{2\lambda+1}}{2\lambda+1}$$

equals $\frac{n^{\theta+1}}{(n+1)^{\lambda+\frac{1}{2}}}$. Therefore, the issue of the tail probability converging to zero is whether $n^{\theta-\lambda+\frac{1}{2}}$ converges to infinity which is the case when $\theta > \lambda - \frac{1}{2}$. The leading order approximation is only valid¹⁸ in the numerator if $\theta > -1$. Therefore, with a negligible loss of the endpoint -1 , the third condition of statistical arbitrage requires $\theta > \max\{\lambda - \frac{1}{2}, -1\}$.

¹⁸For $\theta = -1$, $\sum_{i=1}^n i^{-1}$ is a harmonic series which diverges to infinity with integral representation $\int_1^n s^{-1} ds = \ln(n)$. Therefore, $\theta = -1$ is the critical value which ensures the numerator approaches infinity, a condition required to ensure the probability of a loss approaches zero. Values of θ smaller than -1 result in the numerator converging to a finite number.

The three parameters are tested individually with the Bonferroni¹⁹ inequality accounting for the combined nature of the hypothesis test. The Bonferroni inequality stipulates that the sum of the p values for the individual tests becomes the upper bound for the type I error of the statistical arbitrage tests.

There is no uniformly most powerful test²⁰ as the conditions for applying the Neyman-Pearson Lemma²¹ on composite hypotheses are not satisfied by the non monotonic likelihood function in equation (3). Tests involving joint hypotheses such as a likelihood ratio²² would be biased towards over-rejecting the null hypothesis of no statistical arbitrage. The source of the bias stems from all three conditions on the parameters having to be met in order to generate statistical arbitrage while the likelihood ratio could reject the null if even one of the parameters failed to be in the null set. Furthermore, the null set is the complement of conditions 1-3 in the above theorem, a non standard composite of sets. ■

The point estimates from Lemma 5 need only be tested if they satisfy the properties of Theorem 6. It is well known²³ that maximum likelihood estimation is consistent and asymptotically efficient (achieves Cramer-Rao lower bound). Standard errors for the hypothesis tests in Theorem 6 may be extracted from the Hessian matrix to produce the required t-statistics and their corresponding p values.

As demonstrated above in Theorem 6, based on the distribution of discounted cumulative trading profits found in equation (2), the second and third conditions of statistical arbitrage require $\mu > 0$ and $\theta > \lambda - \frac{1}{2}$, while the fourth condition requires $\lambda < 0$. The condition $\theta > \lambda - \frac{1}{2}$ ensures the probability of a loss converges to zero. The θ parameter may be negative, indicating that expected trading profits are decreasing over time, provided they are not decreasing fast enough to prevent convergence to arbitrage. Thus, the third hypothesis is equivalent to a test for “long run” market efficiency. Observe that the existence of statistical arbitrage is not rejected if θ equals zero, in contrast to the situation when either μ or λ are zero. This leads to a constrained mean test for

¹⁹ $P(\cup_{i=1}^n A_i) \leq \sum_{i=1}^n P(A_i)$, Casella and Berger (1990) page 11.

²⁰ Definition 8.3.5 Casella and Berger (1990) page 365.

²¹ Corollary 8.3.2. Casella and Berger (1990) page 368.

²² Definition 8.2.1 Casella and Berger (1990) page 347.

²³ Casella and Berger (1990) pages 325-326.

statistical arbitrage given in the next subsection.

Furthermore, it is imperative to recognize that applying a single t-statistic to trading profits is not a valid test for statistical arbitrage. Assume λ and θ are both zero, implying the cumulative trading profit $v(t)$ in equation (2) is distributed $\mathcal{N}(\mu n, \sigma^2 n)$. Given an estimate $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\Delta v_i - \hat{\mu})^2$, Theorem 6 reduces to a t-test, with $n - 1$ degrees of freedom, of $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n \Delta v_i$ being statistically positive

$$\hat{\mu} > t_{n-1, \alpha} \sqrt{\frac{\hat{\sigma}^2}{n}}.$$

However, this single t-test is unable to test for statistical arbitrage since the fourth property of statistical arbitrage cannot be verified. In summary, a single t-test²⁴ is limited to only testing the hypothesis of positive expected trading profits which does not imply statistical arbitrage.

Besides the existence of statistical arbitrage, the rate of convergence to arbitrage is also interesting and the subject of the next theorem which states that convergence of the tail probability $P(v(t_n) < 0)$ is faster than exponential. Thus, our introduction of an infinite time horizon is actually less onerous than it appears, as investors quickly realize an investment opportunity that is “close” to arbitrage.

Theorem 7 *The rate of convergence from statistical arbitrage to arbitrage is faster than exponential. Specifically, the convergence rate is faster than*

$$O \left(\exp \left\{ -\frac{\mu^2 \left(\sum_{i=1}^n i^\theta \right)^2}{2\sigma^2 \sum_{i=1}^n i^{2\lambda}} \right\} \right)$$

The proof is contained in Appendix A. Observe that the rate of convergence depends on the squared “Sharpe” ratio $\frac{\mu^2}{\sigma^2}$ as well as the “rates of change” θ and λ . Trading strategies may be evaluated by the convergence rate of $P(v(t_n) < 0)$ to zero and the desirability of trading strategies ranked according to this function.

²⁴The standard t-test performed on the intercept is also unable to test for statistical arbitrage. In addition, these tests assume a particular linear model such as the Fama-French three factor model (1992, 1993) for the model of market equilibrium.

3.1 Unconstrained versus constrained mean statistical arbitrage

The results of this section remain valid when the parameter θ is set to zero, implying the trading strategy's expected profits are constant over time. In this constrained mean model, special cases of Lemma 5 as well as Theorems 6 and 7 immediately result. We refer to this second test as the constrained mean (CM) test for statistical arbitrage to differentiate it from the unconstrained mean (UM) test in Theorem 6.

We begin our empirical analysis with the CM test and later verify that this formulation is appropriate. A variety of robustness checks in Section 6 support the usage of the CM model versus the enhanced UM specification in the context of momentum and value trading strategies. With the parameter $\theta = 0$, the process in equation (1) reduces to

$$\Delta v_i = \mu + \sigma i^\lambda z_i.$$

The corresponding joint hypothesis test for statistical arbitrage is:

$$\text{H1: } \mu > 0,$$

$$\text{H2: } \lambda < 0,$$

while the probability of loss after n periods is equal to

$$\text{Probability of a Loss (after } n \text{ periods)} = N\left(\frac{-\mu n}{\sigma \sqrt{\sum_{i=1}^n i^{2\lambda}}}\right) \quad (8)$$

where $N(\cdot)$ denotes the cumulative standard normal distribution.

4 Data and methodology

Our sample period starts on January 1965 and ends in December 2000. Monthly equity returns data are derived from the Center for Research in Security Prices at the University of Chicago (CRSP). Our analysis covers all stocks traded on the NYSE, AMEX, and NASDAQ that are ordinary common shares (CRSP sharecodes 10 and 11), excluding ADRs, SBIs, certificates, units, REITs, closed-end funds, companies incorporated outside the U.S., and Americus Trust Components.

The stock characteristics data are derived from the CRSP-COMPUSTAT intersection. The stock characteristics we focus on include book-to-market equity, cash flow-to-price ratio, earnings-to-price ratio and annual sales growth. To calculate book-to-market equity, book value per share is taken from the CRSP / COMPUSTAT price, dividends, and earnings database. We treat all negative book values as missing. We take the sum of COMPUSTAT data item 123 (Income before extraordinary items (SCF)) and data item 125 (Depreciation and amortization (SCF)) as cash flow. Only data 123 item is used to calculate cash flow if data 125 item is missing. To compute earnings, we draw on COMPUSTAT data item 58 [Earnings per share (Basic) excluding extraordinary items] and to compute sales we utilize COMPUSTAT data item 12 [sales (net)]. Also, all prices and common shares outstanding numbers employed in the calculation of the ratios are computed at the end of the year.

To ensure that accounting variables are known before hand and to accommodate the variation in fiscal year ends among firms, sorting on stock characteristics is performed in July of year t based on accounting information from year $t-1$. Hence, following Fama and French (1993), to construct the book-to-market deciles from July 1st of year t to June 30th of year $t+1$, stocks are sorted into deciles based on their book-to-market equity (BE/ME), where book equity is book equity in the fiscal year ending in year $t-1$ and market equity is calculated in December of year $t-1$. Similarly, to construct the cash flow-to-price deciles from July 1st of year t to June 30th of year $t+1$, stocks are sorted into deciles based on their cash flow-to-price, where cash flow is cash flow in the fiscal year ending in year $t-1$ and price is the closing price in December of year $t-1$. Earnings-to-price is calculated in a similar fashion. All portfolios are rebalanced every month as some firms disappear from the sample over the 12 month period.

The self-financing condition is enforced by investing (borrowing) trading profits (losses) generated by the various trading strategies in the riskfree asset. Riskfree rate data are obtained from Kenneth French's website.

5 Statistical arbitrage tests of momentum and value strategies

This section presents the results of the tests for statistical arbitrage performed on momentum and value strategies under the assumption that expected incremental trading profits are constant over time. Two hypotheses are jointly tested. First, the incremental profits from the strategy must be statistically greater than zero and second, the time-averaged variance of the strategy must decline to zero as time approaches infinity.

5.1 Momentum strategies

The momentum strategies we focus on are modeled directly after the strategies tested in Jegadeesh and Titman (1993). Specifically we examine strategies that long stocks in the top return decile and short stocks in the bottom return decile. These strategies are based on formation periods of 3, 6, 9, and 12 months and are held for 3, 6, 9, or 12 months.

For example, consider the construction of a momentum strategy with a formation period of three months, and a holding period of six months. Each month, we look back three months to find the top return decile stocks and the bottom return decile stocks. Next, we long the top return decile, short the bottom return decile, and hold this portfolio for six months. The portfolio is rebalanced monthly to account for stocks that drop out of the database.

Given the possible permutations of formation and holding periods, we examine 16 momentum strategies in total. While there are certainly momentum strategies with other formation and holding period combinations that one can examine, doing so might expose our results to a data snooping criticism.

The results of our tests for statistical arbitrage are shown in Table 1. We also present t-ratio tests on the expected profits of the portfolios for comparison. Consistent with Jegadeesh and Titman (1993), the portfolios' expected profits are almost always statistically greater than zero.

Overall, our test for statistical arbitrage confirms the presence of market inefficiency based on several momentum strategies. For 14 of the 16 portfolios, the point estimate for the mean (μ) is greater than zero, and the point estimate for the growth rate of the variance (λ) is less than zero, consistent with statistical arbitrage. Furthermore, at the 5% level, 13 out of 16 momentum trading

strategies have statistically positive estimates for μ . However, the requirement that the time-averaged variance of the incremental profits decline over time appears harder to satisfy. Only 6 of the 16 trading strategies have significantly negative estimates for λ at the 5% level. Nonetheless, the same 6 trading strategies are able to generate statistical arbitrage at the 5% level. Another 3 trading strategies yield statistical arbitrage at the 10% level. In short, roughly half of the momentum strategies examined converge to riskless arbitrages with decreasing time-averaged variances. This finding is hard to reconcile with the notion of market efficiency.

To get a sense of how fast the statistical arbitrages are converging to riskless arbitrages, we plot the probability of a loss, according to Theorem 7, for the three momentum strategies that are statistical arbitrages at the 5% level. The time-averaged variance for the strategies are plotted as well. From Figures 1, 2, and 3, the probability of incurring a loss is comparable across the momentum strategies. For the momentum strategy with a formation period of six months and a holding period of 12 months, after just 86 months, the probability of incurring a loss on this strategy falls below 1%.

Note that strictly speaking, a strategy with a declining probability of loss does not necessarily have a declining time-averaged variance. As an example, consider a probability distribution for trading profits that is moving to the right but whose variance increases at a linear rate or faster.

A casual observation of the time-averaged variance graphs in Figures 1, 2, and 3 reveals that the momentum strategy with a formation period of six months and a holding period of twelve months converges most rapidly to a riskless arbitrage. This observation is corroborated by a comparison of the p values and λ estimates in Table 1.

5.2 Value strategies

Lakonishok, Shleifer, and Vishny (1994) are among the first to examine a comprehensive set of contrarian strategies. They focus on value strategies constructed from stock ratios such as book-to-market, cash flow to price, and earnings to price and on past performance variables like sales growth. They find evidence to suggest that such value strategies are profitable for holding periods ranging from one year to five years. What remains an open question is whether these value strategies

contradict the concept of market efficiency.

In this section, we test whether the strategies examined by Lakonishok, Shleifer, and Vishny (1994) constitute statistical arbitrage opportunities. Specifically we test strategies that long the top decile and short the bottom decile of stocks based on book-to-market, cash flow-to-price, or earnings-to-price ratios of the past year. We hold these portfolios for one, three or five years. We also test a value strategy based on the past three years of sales growth. We long the bottom decile and short the top decile of stocks based on the past three years of sales growth and hold this spread for one, three or five years.

The results in Table 2 suggest that value strategies based on past three years of sales growth are consistent with statistical arbitrage. For all holding periods considered, the sales growth based value strategy tests positively for statistical arbitrage at the 5% level. Value strategies based on cash flow to price also exhibit this tendency. For holding periods of three and five years, the cash flow-to-price based value strategies yield statistical arbitrages at the 5% level. However, their expected profits are lower than those of their sales growth based counterparts. The value strategies based on the other variables do not yield statistical arbitrages at the 5% level. In particular, the point estimates for the variance growth rates of earnings-to-price based strategies (holding period equals one and three years) are positive (though not significantly so) indicating that these strategies may have become riskier over time. Figures 4 and 5 plot the probability of loss and time-averaged variance for two of the value strategies which are statistical arbitrages at the 5% level. Supporting the results of Table 2, both the probability of loss and the time-averaged variance of the sales-based value strategy with a holding period of five years are lower than those of the cash flow to price-based value strategy with a holding period of five years.

Consistent with the results from Fama and French (1992), among others, the expected profits of strategies based on the book-to-market ratio are statistically greater than zero both from the t-ratio test results and from the H1 test (which is joint with H2) results. However, the growth rate of the variance for such strategies is not statistically less than zero, at the 5% level. Book-to-market based value strategies do not constitute statistical arbitrage opportunities. This dovetails with the Fama and French (1993) notion that high book-to-market stocks are more risky than low book-to-market

stocks and hence the former should reward the investor with higher expected profits.

The reason why the cash flow-to-price and earnings-to-price strategies do not perform as well as they do in Lakonishok, Shleifer, and Vishny (1994) may be due to the difference in the sample period. Lakonishok, Shleifer, and Vishny (1994)’s sample period ends in 1989 while our sample period extends till 2000. In results not shown, plots of the cumulative discounted trading profits generated by cash flow-to-price and earnings-to-price based value strategies suggest that these strategies did not perform as well in the post-1990 period as in the pre-1990 period.

Overall, roughly half the value strategies considered (five out of 12) provide evidence against market efficiency. This result is similar to that from testing momentum trading strategies.

6 Robustness checks

The test for statistical arbitrage is performed conditional on a specified process for the discounted incremental trading profits Δv_i . A simple goodness-of-fit test for the assumed incremental trading profit process parallels the standard test conducted on residuals generated by a standard linear regression model. For example, normalized realizations of Δv_i are computed as

$$y_i = \frac{\Delta v_i - \mu i^\theta}{\sigma i^\lambda}$$

and distributed $\mathcal{N}(0,1)$ by assumption. Clearly, y_i values can be generated for other possible mean and variance coefficient combinations in an identical fashion. Thus, normalized trading profit realizations y_i may be tested for normality just as residuals from linear regressions are tested for normality. However, our later simulation analysis demonstrates that in contrast to the sensitivity of existing market efficiency tests to the correct model of market returns, a misspecified process for trading profits biases the test against the presence of statistical arbitrage. Thus, the true nature of our statistical test is very conservative when rejecting market efficiency.

6.1 Simulation study

To gauge the robustness of our assumed process in equation (1), simulations are conducted to investigate the impact of autocorrelation, jumps, and non stationary parameters on inferences

regarding the presence of statistical arbitrage. Using parameters μ , σ , θ , and λ consistent with no statistical arbitrage and statistical arbitrage, a time series of 400 incremental trading profits are simulated. Jumps are simulated from a normal distribution while non stationary parameters are modeled using a two state Markov chain with low and high values for μ and σ that simulate incremental trading profits deviating²⁵ from Assumption 1. Statistical arbitrage opportunities are simulated using parameters $\mu = .01$, $\sigma = .02$, $\theta = -.05$, and $\lambda = -.15$ while trading profits that are not consistent with statistical arbitrage are generated by parameters $\mu = .01$, $\sigma = .02$, $\theta = -.05$, and $\lambda = 0$. Different values for λ are considered with the statistical procedure performing *perfectly* whenever $\lambda > 0$. Setting $\mu \leq 0$ produced similar results. The critical threshold of $\lambda = 0$ is chosen to make the test for statistical arbitrage as difficult as possible with μ remaining at .01. The ϕ parameter used to generate autocorrelation following the MA(1) process specified in Assumption 2 of Appendix C is .40.

Secondly, estimation involving Lemma 5 based on the assumed process in equation (1) is then conducted on each simulated time series and a decision rendered as to whether the simulated trading profits are consistent with statistical arbitrage. This step involved summing the p values for the various hypothesis tests and rejecting statistical arbitrage if the sum exceeded .05 (or .10). A total of 100 time series are simulated with parameter estimation following each simulation.

The intensity parameter for jumps is chosen to yield an average of 1 jump every 3 years with jump magnitudes normally distributed with zero mean and standard deviation equal to .05. A two state Markov chain generated non stationary drift and volatility parameters according to the transition matrix

$$\begin{bmatrix} .8 & .2 \\ .2 & .8 \end{bmatrix}$$

for low and high μ and σ values. The low and high values for μ are .005 and .01, respectively, while σ had low and high values of .015 and .03. The parameter value for θ remained at $-.05$. In addition, λ values of $-.15$ and 0 are employed to simulate circumstances with and without

²⁵No null hypothesis in relation to statistical arbitrage for the jump or Markov chain parameters is required. We are only interested in studying the impact of jumps and non stationarity on inferences based on the assumed process.

statistical arbitrage. Thus, the λ parameter is responsible for differentiating between economies with and without statistical arbitrage as in the previous three simulations.

The number of conclusions that are altered by autocorrelation, jumps, or non stationarity are recorded in Table 3. Observe that the statistical test correctly accepts the null hypothesis of no statistical arbitrage with great accuracy, even when there exists a deviation from the assumed process. At the .10 level, when the underlying parameters are consistent with no statistical arbitrage, the test rejects the hypothesis no more than 5% of the time.

Overall, the “power” of the statistical test is exceptional even with deviations in the assumed process. The simulations imply that the presence of autocorrelation, jumps, and parameter non stationarity lead to a bias towards accepting the null hypothesis of no statistical arbitrage. Hence, the formulation in equation (1) may be considered “fail-safe” in the presence of deviations. This property stems from the fourth property of statistical arbitrage which requires λ to be negative. The additional volatility caused by jumps and non stationary parameters increases the standard error of λ . This translates into higher corresponding p values for tests involving λ , and consequently a higher probability of accepting the null hypothesis of no statistical arbitrage. Along with the Bonferroni inequality, these simulations demonstrate that our test for statistical arbitrage is quite conservative when rejecting the null hypothesis of no statistical arbitrage. In summary, our simulation analysis reveals that a statistical arbitrage opportunity allows one to confidently reject market efficiency.

6.2 Testing the constrained mean model

Our second robustness check verifies whether the constrained mean (CM) model, which assumes expected trading profits are constant over time, offers a good fit for the incremental profit processes we examine. To do so, we compare measures of fit for the CM model with those of the unconstrained mean (UM) model. We also study the estimated rates of change in the expected trading profits to determine whether they are significantly negative. From a statistical perspective, introducing the θ parameter, whose estimates are usually negative, reduces the number of significantly positive μ estimates. Overall, for the UM test, the point estimates of μ increase along with their standard errors as the information contained in the trading profits is spread over a fourth parameter.

The first measure of fit we examine is the average root mean squared error (RMSE). The RMSE numbers for each model are calculated by comparing the portfolio’s observed trading profit and those of 100 simulations, based on the estimated parameters of the observed trading profit. If the UM model offers a better fit of the data than the CM model, then the RMSE numbers for the former should be lower than those of the latter. However, Table 4, which records the RMSE numbers for both models, demonstrates that the UM model does not offer a better statistical alternative than its CM counterpart. In fact, the RMSE numbers are almost indistinguishable.

The second measure of fit we look at is the sum of normalized squared residuals (SSR). If the SSR numbers of the UM model are lower than those of the CM model, then we are lead to conclude that it is the better model for the data. Yet Table 4 shows that the SSR numbers for the two models are again almost identical.

Next, we test whether the incremental profits of the portfolios are decreasing using a t-statistic. The p-values for this test are also recorded in Table 4. We find that for all except one²⁶ of the portfolios which are CM statistical arbitrages at the 10% or 5% level, the growth rate of the incremental profits is statistically indistinguishable from zero. Therefore, there is no need to estimate θ and weaken the power of the statistical arbitrage test.

Overall, the UM test for statistical arbitrage unnecessarily increases the standard errors of the mean incremental profit estimates without offering a better goodness of fit as compensation. Hence the CM model is more appropriate for modeling the incremental trading profits we observe.

6.3 Transactions costs

A common critique of financial anomalies is that the trading profits from such anomalies disappear after adjusting for transactions costs. In this section we outline a procedure to estimate the transactions costs of the portfolios in Section 4. We then re-test the portfolios for statistical arbitrage after adjusting the incremental profits downward by their respective transactions costs.

Our procedure for estimating transactions costs is as follows. First, we estimate the average

²⁶Interestingly, this portfolio (MOM 9/12) is a statistical arbitrage according to the UM test. Therefore, for the one instance where trading profits are better modeled using a UM process with a significantly negative θ estimate, the same conclusion is reached regarding market efficiency.

monthly turnover for each of the portfolios by taking a ratio of the sum of buys and sells each period over two times the total number of stocks held in that period. This estimates the number of round trip transactions as a percentage of the number of stocks held. Then we multiply this measure of monthly turnover with an estimate of the round trip transactions cost given in Chan and Lakonishok (1997). According to Chan and Lakonishok (1997), the average complexity stock in the NASDAQ has a round trip transactions cost²⁷ of 1.34%.

Finally we adjust the monthly profits downward by the transactions costs. The statistical arbitrage test from Section 2 is then re-applied. The results in Table 5 suggest that all of the 5% statistical arbitrage portfolios survive the adjustment for transactions costs. Nine of the 11 portfolios remain statistical arbitrages at the 5% level after accounting for transactions costs, while two of the remaining portfolios yield statistical arbitrages at the 10% level. This suggests that the statistical arbitrages represent feasible and attractive trading opportunities for market participants.

6.4 Influence of small stocks

Several tests of stock market anomalies conclude that small stocks are more inefficient than large stocks (Hong, Lim, and Stein, 2000; Mitchell and Stafford, 2000). Hence, it may be interesting to see if small stocks are driving our results. In addition, it is well-known that small stocks are less liquid, and have greater transactions costs than large stocks (Chan and Lakonishok, 1997). So it may be useful to see if our results survive the exclusion of these illiquid stocks.

To this end, we remove stocks in our sample with market capitalizations below the 40th NYSE market equity percentile. This reduces our sample size by more than half since there are many more small stocks than large stocks and there are relatively more small stocks traded in the NASDAQ (which are included in the CRSP database) than in the NYSE. We then re-test for statistical arbitrage.

Results from the tests of statistical arbitrage suggest that evidence of statistical arbitrage is even greater without the bottom 40th NYSE percentile. All the momentum strategies and 10 out of

²⁷We also calculate transactions costs using the higher NYSE estimate of 1.55% and re-test for statistical arbitrage. None of the inferences change with the NYSE estimate for round trip transactions cost.

12 of the value strategies are statistical arbitrages at the 5% level²⁸. The sales based value strategy with a holding period of 5 years tests positively for a statistical arbitrage at the 10% level. Only the sales-based value strategies with a holding period of 3 fails to be a statistical arbitrage. This clearly indicates that most of the statistical arbitrages are not driven by small stocks.

6.5 Final comments on robustness

One point to emphasize is the conservative nature of our statistical test. Despite the accuracy of our estimation procedure, the Bonferroni inequality results in a conservative test of the joint hypothesis for statistical arbitrage since the sum of p values must be below the α level. In addition, deviations from our assumed process bias the test against rejecting the null hypothesis of no statistical arbitrage. Hence, evidence of statistical arbitrage is not a consequence of a misspecified trading profit process. In the context of this paper, only autocorrelation in the trading profits of the momentum strategies appears to be of any consequence.

For completeness and potential applications to other trading strategies in future research, we provide a brief summary of enhanced statistical tests that account for autocorrelated trading profits in Appendix C. Empirically implementing this material did not alter our conclusions regarding market efficiency or offer any additional economic insights²⁹.

7 Conclusion

This paper introduces the concept of statistical arbitrage which facilitates a test of market efficiency without the need to specify an equilibrium model. In the limit, statistical arbitrage converges to arbitrage. Consequently, the joint hypothesis dilemma is avoided by appealing to a long horizon trading strategy. Methodologies that test for the existence of statistical arbitrage are provided. Simulations reveal the robustness and the conservative nature of our statistical test, with a bias towards accepting the null hypothesis of no statistical arbitrage in the presence of deviations from our assumed trading profit process such as autocorrelation, jumps, and non stationary parameters.

²⁸The results are available upon request.

²⁹The results are also available upon request.

Tests for statistical arbitrage are applied to the cumulative discounted trading profits of momentum (Jegadeesh and Titman, 1993) and value trading strategies (Lakonishok, Shleifer, and Vishny, 1994). In contrast to the t-test of positive “risk-adjusted” expected profits, the existence of statistical arbitrage is sufficient to reject market efficiency for any model of market returns.

Roughly half of the momentum and value strategies we evaluate test positively for statistical arbitrage. This suggests the existence of several trading opportunities that converge to riskless arbitrages with decreasing time-averaged variances, a result difficult to reconcile with market efficiency. Our test results are robust to the incorporation of transactions costs and the exclusion of small stocks.

Promising avenues of research include testing other anomalies for statistical arbitrage, such as the abnormal returns from dividends or earnings announcements. Also, the Bonferroni inequality may be replaced by a more computationally intensive monte carlo procedure in order to simulate the critical values underlying our joint hypothesis test. However, preliminary evidence suggests that the monte carlo methodology does not alter our conclusions regarding market efficiency.

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Appendices

A Zero probability of loss in limit

Equation (2) states that $v(t_n)$ is distributed

$$\mathcal{N}\left(\mu \sum_{i=1}^n i^\theta, \sigma^2 \sum_{i=1}^n i^{2\lambda}\right).$$

Convergence of the tail probability, $P(v(t_n) < 0)$, to zero is determined by the cumulative normal distribution

$$P(v(t_n) < 0) = \frac{1}{\sqrt{2\pi\sigma^2 \sum_{i=1}^n i^{2\lambda}}} \int_{-\infty}^0 e^{-\frac{1}{2\sigma^2 \sum_{i=1}^n i^{2\lambda}}(x - \mu \sum_{i=1}^n i^\theta)^2} dx.$$

The above probability may be expressed as

$$P(v(t_n) < 0) = P\left(Z < \frac{-\mu \sum_{i=1}^n i^\theta}{\sigma \sqrt{\sum_{i=1}^n i^{2\lambda}}}\right) = N\left(\frac{-\mu \sum_{i=1}^n i^\theta}{\sigma \sqrt{\sum_{i=1}^n i^{2\lambda}}}\right)$$

where Z is a standard normal and $N(x)$ is the standard normal cumulative distribution function. Although there is no closed form solution for the standard normal cdf, a polynomial approximation (for $x < 0$) is available, Hull (1997) page 243

$$N(x) = N'(x) \left(a_1 \frac{1}{1 + \gamma x} + a_2 \frac{1}{(1 + \gamma x)^2} + a_3 \frac{1}{(1 + \gamma x)^3} + \text{h.o.t.} \right)$$

where $a_1 = 0.4361836$, $a_2 = -0.1201676$, $a_3 = 0.9372980$ and $\gamma = .33267$ are constants. The leading term governs the rate of convergence implying

$$P(v(t_n) < 0) = O\left(\frac{a_1}{\sqrt{2\pi}} \exp\left\{-\frac{\mu^2 (\sum_{i=1}^n i^\theta)^2}{2\sigma^2 \sum_{i=1}^n i^{2\lambda}}\right\} \frac{1}{1 + \gamma \frac{\mu \sum_{i=1}^n i^\theta}{\sigma \sqrt{\sum_{i=1}^n i^{2\lambda}}}}\right)$$

which converges to zero as n increases under the assumption of statistical arbitrage. In particular, the probability converges faster than rate

$$O\left(\exp\left\{-\frac{\mu^2 (\sum_{i=1}^n i^\theta)^2}{2\sigma^2 \sum_{i=1}^n i^{2\lambda}}\right\}\right)$$

as a function of n .

B Importance of fourth condition

Consider the standard Black-Scholes economy with

$$S_t = S_0 e^{\alpha t - \sigma^2 t/2 + \sigma W_t}$$

and assume $r = 0$ implying $B_t \equiv 1$ for all t . This economy is in equilibrium but would be rejected by a definition of statistical arbitrage that did not include the fourth condition. To prove this, first consider a simpler economy with stock dynamics

$$\bar{s}_t = \bar{s}_0 + \alpha t - \sigma^2 t/2 + \sigma W_t$$

and a buy and hold strategy similar to Example 2 whose value at time t equals $\bar{v}(t) = 1 \cdot \bar{s}_t - \bar{s}_0 \cdot 1$. For $r = 0$, cumulative trading profits are identical to discounted cumulative trading profits. Cumulative trading profit dynamics are represented as

$$\begin{aligned} \bar{v}(t) &= \bar{s}_0 + \alpha t - \sigma^2 t/2 + \sigma W_t - \bar{s}_0 \\ &= [\alpha - \sigma^2/2]t + \sigma W_t. \end{aligned}$$

If $\alpha - \sigma^2/2 > 0$, then $P(\bar{v}(t) < 0)$ converges to zero. This claim follows from Appendix A with λ and θ both equaling zero and $\mu = \alpha - \sigma^2/2$. This portfolio would also satisfy conditions 1 and 2 of statistical arbitrage. Returning to the Black-Scholes economy with $S_0 = 1$, consider the same buy and hold trading strategy with cumulative trading profits

$$v(t) = 1 \cdot S_t - S_0 \cdot 1 = S_t - 1 = e^{\bar{s}_t} - 1.$$

The following sequence illustrates that $P(v(t) < 0)$ also converges to zero

$$P(v(t) < 0) = P(e^{\bar{s}_t} - 1 < 0) < P(\bar{s}_t - 1 < 0) = P(\bar{v}(t) < 0)$$

where the second relation employs the inequality $S_t = e^{\bar{s}_t} > \bar{s}_t$. Thus, without condition 4, the definition of statistical arbitrage would apply to a buy and hold strategy in the Black-Scholes economy which is known to be in equilibrium.

C Correlated trading profit increments

Trading strategies may produce autocorrelated trading profits. For example, the momentum trading strategies have profits derived from overlapping periods. Consider an $MA(1)$ process to instill correlation between consecutive³⁰ trading profit increments according to the parameter ϕ . Specifically, the correlation between Δv_i and its previous value Δv_{i-1} is approximately $\frac{\phi}{1+\phi^2}$ under the following assumption.

Assumption 2 *Let the discounted incremental trading profits satisfy*

$$\Delta v_i = \mu i^\theta + i^\lambda z_i + \phi(i-1)^\lambda z_{i-1}$$

for $i = 1, 2, \dots, n$ where z_i are i.i.d $\mathcal{N}(0, \sigma^2)$ random variables with $z_0 = 0$.

For parameter estimation, several iterative procedures exist as detailed in Box and Jenkins (1976) chapter 7. The distribution of $v(t_n)$ is required to construct a test for statistical arbitrage and determine the convergence rate to arbitrage. The distribution of the cumulative discounted trading profits is the topic of the next lemma.

Lemma 8 *The distribution of $v(t_n)$ with correlated increments equals*

$$v(t_n) = \mathcal{N}\left(\mu \sum_{i=1}^n i^\theta, \sigma^2 \left[(1+\phi)^2 \sum_{i=1}^{n-1} i^{2\lambda} + n^{2\lambda} \right]\right). \quad (9)$$

Proof. The distribution of $v(t_n)$ is solved by recursion

$$\begin{aligned} v(t_i) &= v(t_{i-1}) + \mu i^\theta + i^\lambda z_i + \phi(i-1)^\lambda z_{i-1} \\ &= v(t_{i-2}) + \mu(i-1)^\theta + (i-1)^\lambda z_{i-1} + \phi(i-2)^\lambda z_{i-2} \\ &\quad + \mu i^\theta + i^\lambda z_i + \phi(i-1)^\lambda z_{i-1} \\ &= v(t_{i-3}) + \mu(i-2)^\theta + (i-2)^\lambda z_{i-2} + \phi(i-3)^\lambda z_{i-3} \\ &\quad + \mu(i-1)^\theta + (i-1)^\lambda z_{i-1} + \phi(i-2)^\lambda z_{i-2} \\ &\quad + \mu i^\theta + i^\lambda z_i + \phi(i-1)^\lambda z_{i-1} \end{aligned}$$

³⁰Generalizing the process to an $MA(q)$ for $q \geq 2$ follows in a straightforward manner.

and so forth generating the following solution for $v(t_n)$

$$\begin{aligned} v(t_n) &= \mu \sum_{i=1}^n i^\theta + (1 + \phi) \sum_{i=1}^{n-1} i^\lambda z_i + n^\lambda z_n \\ &= \mathcal{N} \left(\mu \sum_{i=1}^n i^\theta, \sigma^2 \left[(1 + \phi)^2 \sum_{i=1}^{n-1} i^{2\lambda} + n^{2\lambda} \right] \right). \end{aligned} \quad (10)$$

■

As expected, equation (9) reduces to equation (2) when $\phi = 0$.

Theorem 9 *Under the previous assumption, a trading strategy generates a statistical arbitrage with $1 - \alpha$ percent confidence if the following conditions are satisfied:*

H1: $\mu > 0$,

H2: $\lambda < 0$,

H3: $\theta > \max \left\{ \lambda - \frac{1}{2}, -1 \right\}$,

with the sum of the p values for the individual tests forming an upper bound for the type I error α .

Proof. The variance in equation (10) may be modified to equal $\sigma^2(1 + \phi)^2 \sum_{i=1}^n i^{2\lambda}$ by adding the term $\sigma^2(2\phi + \phi^2)n^{2\lambda}$. The marginal contribution of this additional term

$$\frac{\sigma^2 \left[(1 + \phi)^2 \sum_{i=1}^{n-1} i^{2\lambda} + n^{2\lambda} \right] + \sigma^2(2\phi + \phi^2)n^{2\lambda}}{\sigma^2 \left[(1 + \phi)^2 \sum_{i=1}^{n-1} i^{2\lambda} + n^{2\lambda} \right]} \rightarrow 0$$

as n increases, ensuring asymptotic inferences remain valid. Consequently, for large n , the approximation $v(t_n) \stackrel{d}{\sim} \mathcal{N} \left(\mu \sum_{i=1}^n i^\theta, \sigma^2(1 + \phi)^2 \sum_{i=1}^n i^{2\lambda} \right)$ is analyzed. Therefore, testing for statistical arbitrage involves the same exact joint hypotheses as Theorem 6 but with a modified variance that accounts for ϕ . ■

The next theorem follows immediately from Theorem 7.

Theorem 10 *The rate of convergence from statistical arbitrage to arbitrage is exponential with respect to n . Specifically, the convergence rate is faster than*

$$O \left(\exp \left\{ - \frac{\mu^2 \left(\sum_{i=1}^n i^\theta \right)^2}{2\sigma^2(1 + \phi)^2 \sum_{i=1}^n i^{2\lambda}} \right\} \right)$$

Theorems 9 and 10 illustrate that, conditional on parameters μ , σ^2 , θ , and λ being equal, positive correlation implies slower convergence (higher variance) with $(1 + \phi)^2 > 1$ while negative correlation implies faster convergence (lower variance) with $(1 + \phi)^2 < 1$.

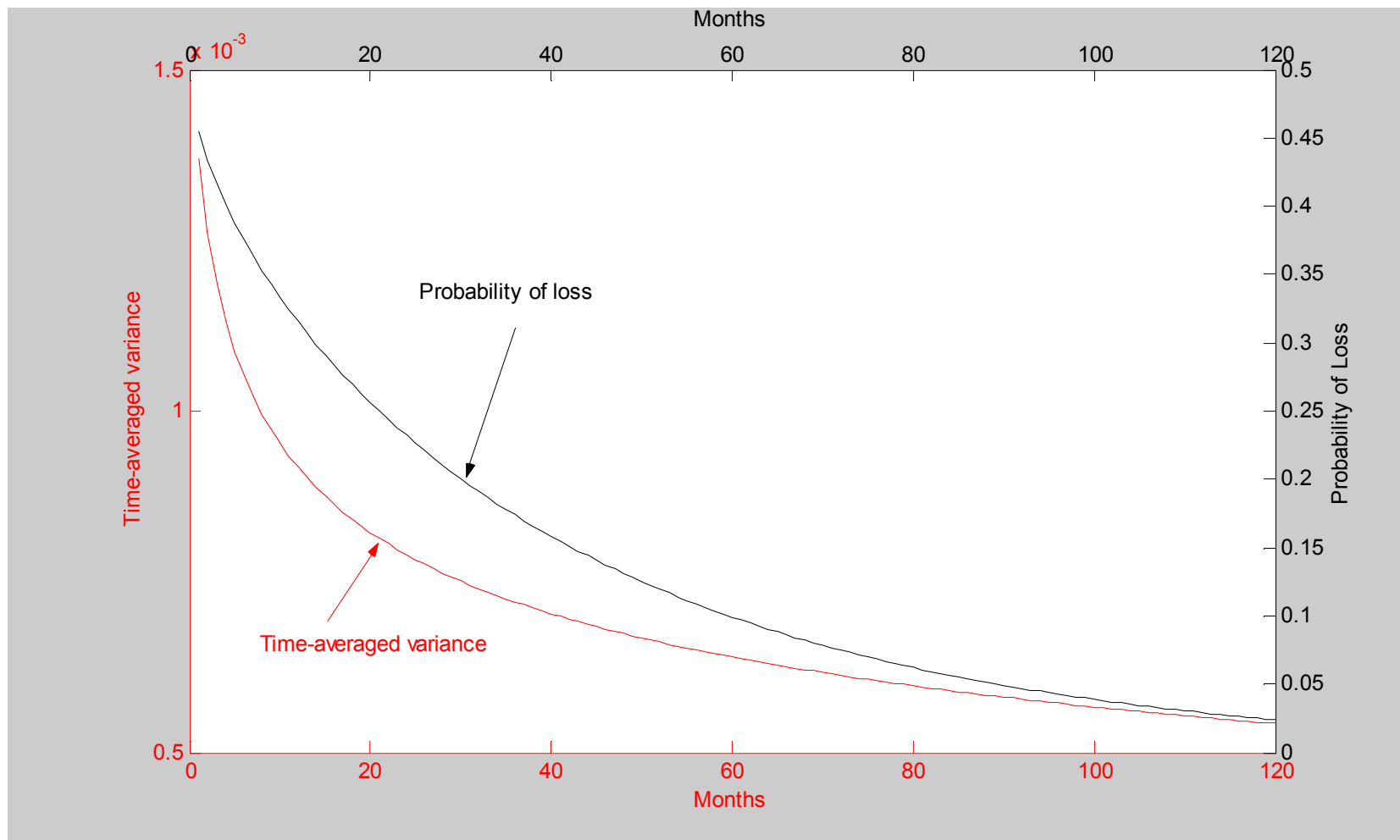


Fig 1. Probability of a loss and time-averaged variance for the momentum strategy (6,6) which represents a CM statistical arbitrage opportunity. Sample period is from January 1965 to December 2000. The momentum strategy (6,6) longs the top return decile of stocks and shorts the bottom return decile of stocks in the sample based on a formation period of 6 months and a holding period of 6 months. The probability of loss is calculated as per equation (8).

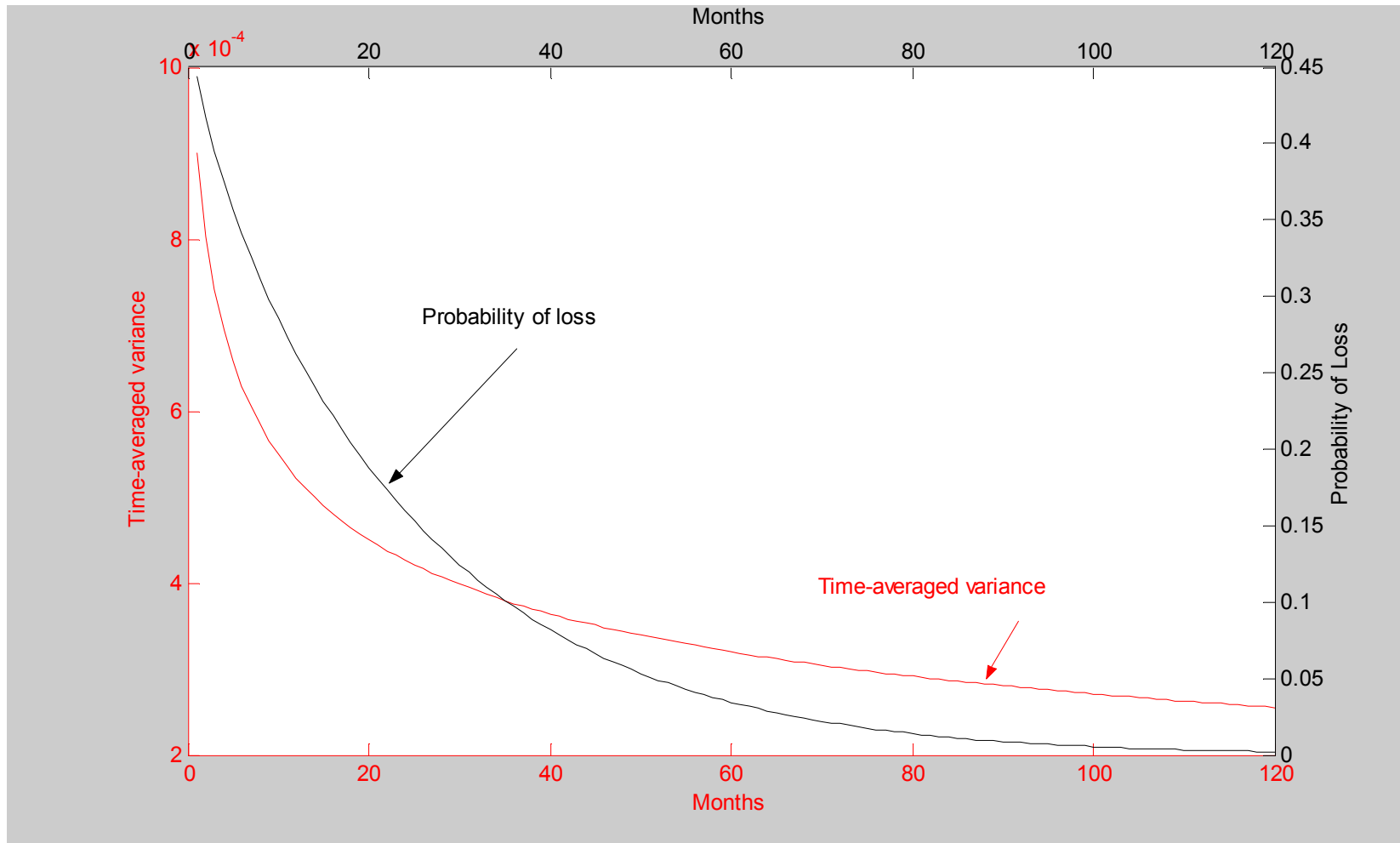


Fig 2. Probability of a loss and time-averaged variance for the momentum strategy (6,12) which represents a CM statistical arbitrage opportunity. Sample period is from January 1965 to December 2000. The momentum strategy (6,12) longs the top return decile of stocks and shorts the bottom return decile of stocks in the sample based on a formation period of 6 months and a holding period of 12 months. The probability of loss is calculated as per equation (8).

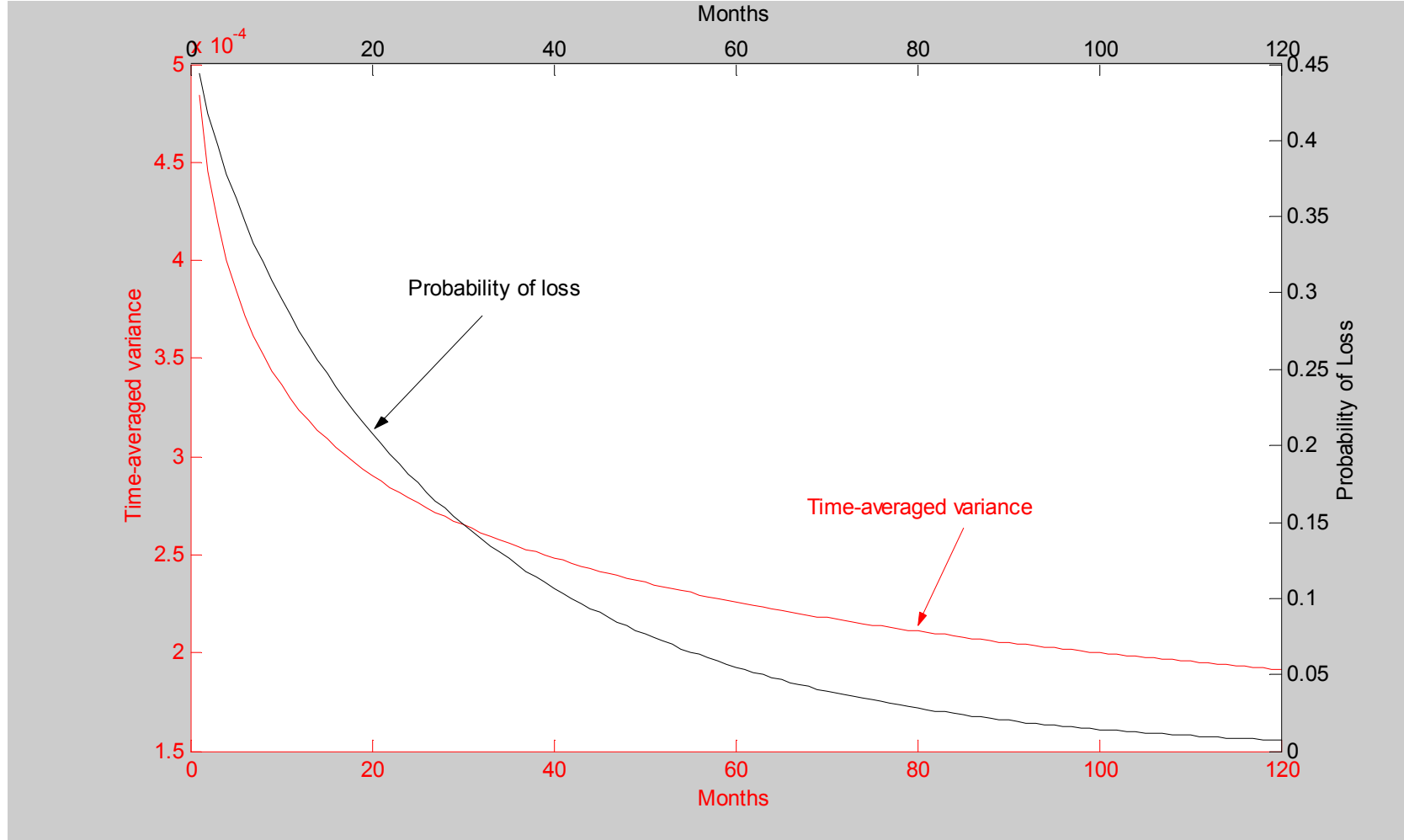


Fig 3. Probability of a loss and time-averaged variance for the momentum strategy (9,12) which represents a CM statistical arbitrage opportunity. Sample period is from January 1965 to December 2000. The momentum strategy (9,12) longs the top return decile of stocks and shorts the bottom return decile of stocks in the sample based on a formation period of 9 months and a holding period of 12 months. The probability of loss is calculated as per equation (8).

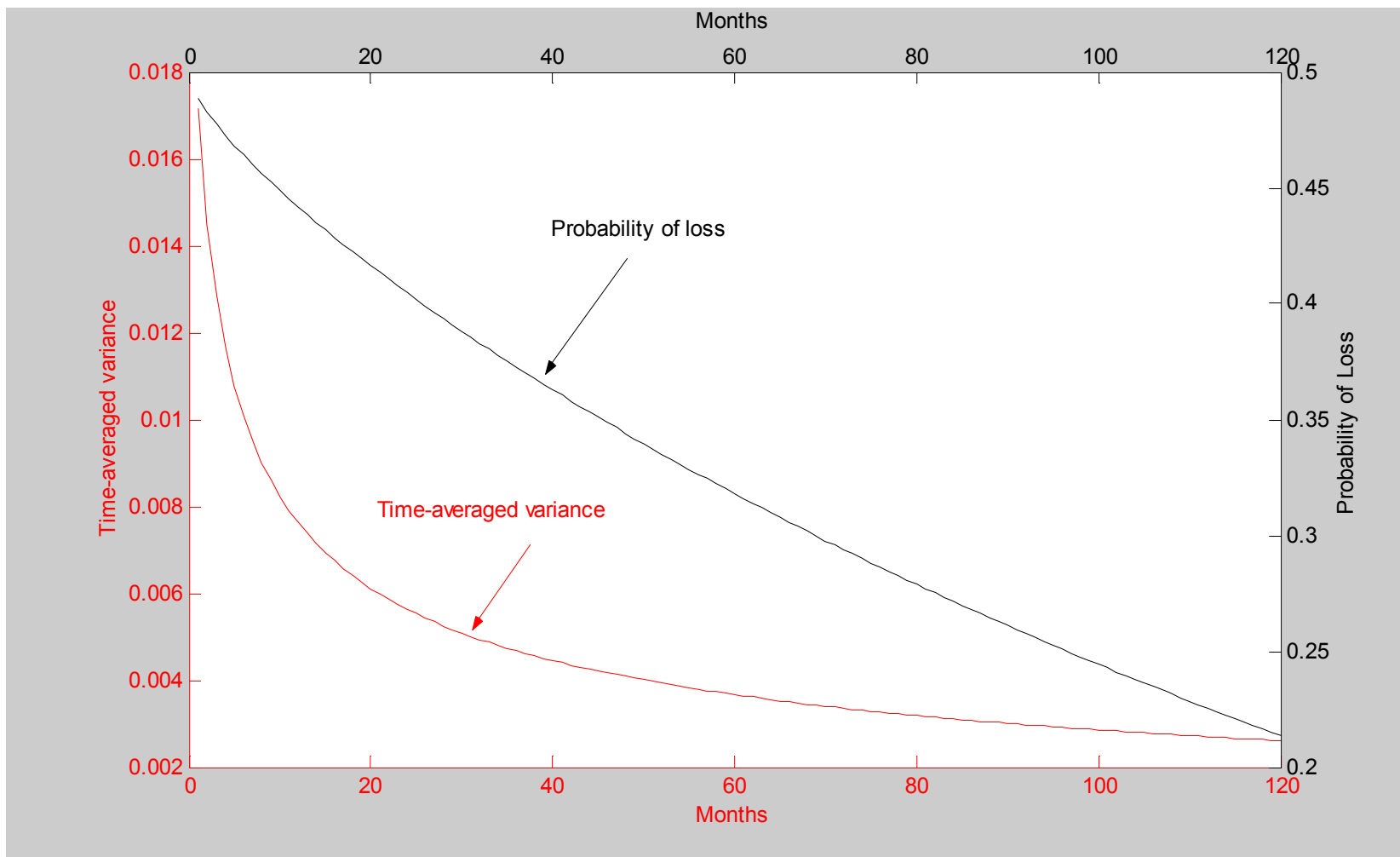


Fig 4. Probability of a loss and time-averaged variance for the value strategy CP/5 which represents a CM statistical arbitrage opportunity. Sample period is from January 1965 to December 2000. The value strategy CP/5 longs the bottom past year cash flow to price decile of stocks and shorts the top past year cash flow to price decile of stocks in the sample and holds the spread for 5 years. The probability of loss is calculated as per equation (8).

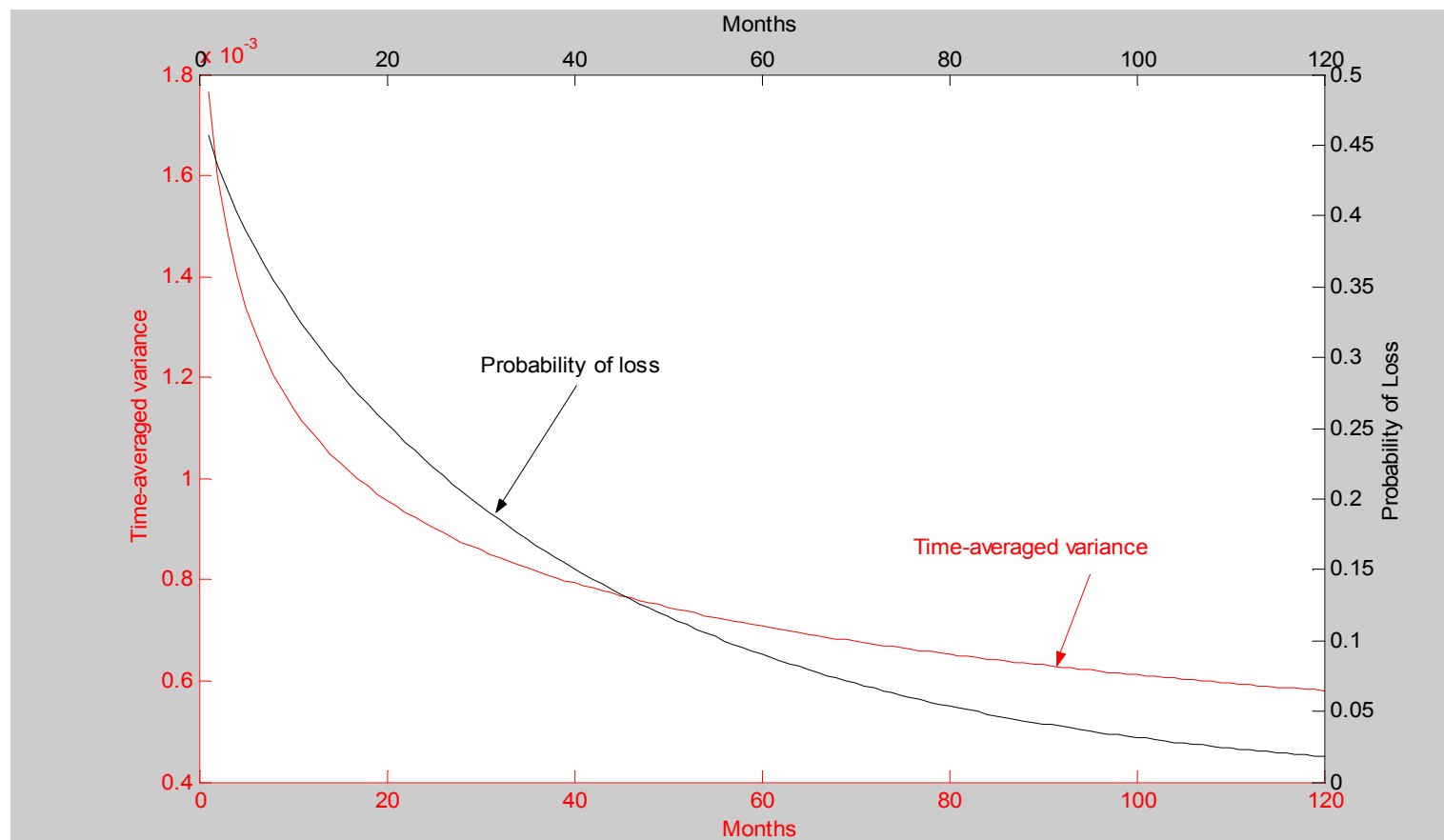


Fig 5. Probability of a loss and time-averaged variance for the value strategy Sales/5 which represents a CM statistical arbitrage opportunity. Sample period is from January 1965 to December 2000. The value strategy Sales/5 longs the bottom past 3-year sales growth decile of stocks and shorts the top past 3-year sales growth decile of stocks in the sample and holds the spread for 5 years. The probability of loss is calculated as per equation (8).

Table 1

Constrained Mean (CM) Statistical Arbitrage Tests: Jegadeesh and Titman (1993) Momentum Strategies

Parameter estimates and corresponding p values for constrained mean test of statistical arbitrage. Sample period is from January 1965 to December 2000. Portfolio x / y denotes a momentum portfolio with a formation period of x months and a holding period of y months. Every month, stocks are sorted based on past x months of returns into decile portfolios. The portfolio x / y longs the top decile and shorts the bottom decile and holds that spread for y months as in Jegadeesh and Titman (1993). The risk free asset is used to finance the portfolio. H1 and H2 denote the p values from statistical arbitrage tests which test whether the portfolio's mean monthly incremental profit is positive and whether its time-averaged variance is declining over time. The sum of the H1 and H2 columns is the p value for the statistical arbitrage test. Note that the sum may exceed one because of the Bonferonni inequality. The t-statistic on the mean monthly trading profit is provided for comparison.

Portfolio	μ (mean)	t-stat	σ (std dev)	λ (growth rate of std dev)	H1 ($\mu > 0$)	H2 ($\lambda < 0$)	Sum (H1+H2)	Statistical Arbitrage
MOM 3/3	-0.0027	-1.78	0.033	-0.0174	0.967	0.306	1.273	No
MOM 3/6 [^]	0.0015	1.68	0.023	-0.035	0.056	0.156	0.212	No
MOM 3/9 [^]	0.0021	3.08	0.025	-0.1083	0.001	0.001	0.002	Yes**
MOM 3/12 [^]	0.0033	5.81	0.023	-0.1317	0.000	0.000	0.000	Yes**
MOM 6/3 [^]	0.002	1.27	0.037	-0.0168	0.11	0.306	0.416	No
MOM 6/6 [^]	0.0042	4.45	0.037	-0.1254	0.000	0.000	0.000	Yes**
MOM 6/9 [^]	0.0055	6.78	0.026	-0.0908	0.000	0.004	0.004	Yes**
MOM 6/12 [^]	0.0042	6.92	0.03	-0.1734	0.000	0.000	0.000	Yes**
MOM 9/3 [^]	0.0035	2.24	0.043	-0.0532	0.013	0.075	0.088	Yes*
MOM 9/6 [^]	0.0065	6.53	0.027	-0.0603	0.000	0.052	0.052	Yes*
MOM 9/9 [^]	0.0053	6.93	0.021	-0.0549	0.000	0.064	0.064	Yes*
MOM 9/12 [^]	0.0031	5.78	0.022	-0.1256	0.000	0.001	0.001	Yes**
MOM 12/3 [^]	0.006	3.93	0.035	-0.0206	0.000	0.266	0.266	No
MOM 12/6 [^]	0.0057	6.13	0.02	-0.01	0.000	0.385	0.385	No
MOM 12/9	0.0038	5.19	0.012	0.0432	0.000	0.878	0.878	No
MOM 12/12	0.0015	3.00	0.01	-0.0004	0.002	0.496	0.498	No

* Significant at the 10% level

** Significant at the 5% level

[^] Point estimates of μ and λ consistent with statistical arbitrage

Table 2

Constrained Mean (CM) Statistical Arbitrage Tests: Lakonishok, Shleifer, and Vishny (1994) Value Strategies

Parameter estimates and corresponding p values for constrained mean test of statistical arbitrage. Sample period is from January 1965 to December 2000. As in Lakonishok, Shleifer, and Vishny (1994), the sorting variables are past one-year book-to-market (BM), cash flow-to-price (CP), and earnings-to-price (EP) as well as past three-year sales growth (SALES). Portfolio x / y denotes a value portfolio based on stock characteristic x and a holding period of y years. Every July, stocks are sorted into decile portfolios based on a particular stock characteristic. For BM, CP, and EP portfolios, the portfolio x / y longs the top decile and shorts the bottom decile and holds that spread for y years. For SALES portfolios, the portfolio x / y longs the bottom decile and shorts the top decile and holds that spread for y years. The risk free asset is used to finance the portfolio. H1 and H2 denote the p values from statistical arbitrage tests which test whether the portfolio's mean monthly incremental profit is positive and whether its time-averaged variance is declining over time. The sum of the H1 and H2 columns is the p value for the statistical arbitrage test. Note that the sum may exceed one because of the Bonferonni inequality. The t-statistic on the mean monthly trading profit is provided for comparison.

Portfolio	μ (mean)	t-stat	σ (std dev)	λ (growth rate of std dev)	H1 ($\mu > 0$)	H2 ($\lambda < 0$)	Sum (H1+H2)	Statistical Arbitrage
BM/1^	0.0141	6.34	0.054	-0.0373	0.000	0.174	0.174	No
BM/3^	0.0115	6.54	0.042	-0.0427	0.000	0.166	0.166	No
BM/5^	0.0106	6.46	0.041	-0.0720	0.000	0.063	0.063	Yes*
CP/1^	0.0021	1.28	0.196	-0.2179	0.249	0.000	0.249	No
CP/3^	0.0036	1.74	0.132	-0.2272	0.046	0.000	0.046	Yes**
CP/5^	0.0037	1.39	0.131	-0.2685	0.027	0.000	0.027	Yes**
EP/1	0.0021	0.25	0.015	0.2320	0.177	1.000	1.177	No
EP/3	0.0013	0.57	0.026	0.0735	0.246	0.980	1.226	No
EP/5^	0.0011	0.63	0.039	-0.0494	0.268	0.079	0.347	No
SALES/1^	0.0085	5.79	0.043	-0.0856	0.000	0.019	0.019	Yes**
SALES/3^	0.0056	4.65	0.036	-0.0936	0.000	0.028	0.028	Yes**
SALES/5^	0.0046	3.83	0.042	-0.1517	0.000	0.001	0.001	Yes**

* Significant at the 10% level

** Significant at the 5% level

^ Point estimates of μ and λ consistent with statistical arbitrage

Table 3**Simulation tests at the 5% and the 10% levels**

This table reports simulation results on the accuracy of the statistical arbitrage test in the face of deviations from the assumed incremental profit process. The first column details the departure from the assumed dynamics in equation (1). The second column indicates whether or not the original parameters are consistent with statistical arbitrage, based on λ being either 0.15 or 0. Columns three and four record the number of instances in the 100 simulated time series that the hypothesis of no statistical arbitrage is correctly rejected or accepted at the 5% and the 10% levels.

Type of deviation	Statistical Arbitrage	Accuracy .05 Level	Accuracy .10 Level
No Deviation	Yes	99%	100%
Autocorrelation	Yes	85%	98%
Jumps	Yes	58%	67%
Non Stationary μ, σ	Yes	65%	81%
No Deviation	No	100%	99%
Autocorrelation	No	100%	97%
Jumps	No	99%	95%
Non Stationary μ, σ	No	100%	95%

Table 4
Comparison Between Constrained Mean (CM) and Unconstrained Mean (UM)
Models

A comparison of the Root Mean Squared Errors (RMSE) and Sum of Squared Residuals (SSR) between constrained mean and unconstrained mean models of statistical arbitrage for the momentum portfolios and value portfolios in Tables 1 and 2. The p value for the test that the incremental trading profit of the portfolios is declining over time is also presented. The RMSE is based on a monte carlo experiment with 100 simulated portfolios.

Portfolio	RMSE (CM)	RMSE (UM)	SSR (CM)	SSR (UM)	p-value, θ (growth rate of mean) <0
MOM 3/3	0.879	0.879	1.000	1.000	0.83
MOM 3/6	0.555	0.555	1.001	1.000	0.00
MOM 3/9**	0.435	0.430	1.003	1.003	0.14
MOM 3/12**	0.349	0.350	1.004	1.004	0.20
MOM 6/3	0.984	0.980	1.000	1.000	0.00
MOM 6/6**	0.583	0.576	1.003	1.003	0.38
MOM 6/9**	0.479	0.477	1.002	1.002	0.41
MOM 6/12**	0.371	0.370	1.005	1.004	0.25
MOM 9/3*	0.950	0.944	1.001	1.001	0.17
MOM 9/6*	0.587	0.585	1.002	1.002	0.53
MOM 9/9*	0.442	0.448	1.001	1.001	0.32
MOM 9/12**	0.332	0.332	1.003	1.003	0.01
MOM 12/3	0.908	0.907	1.001	1.001	0.07
MOM 12/6	0.539	0.545	1.000	1.000	0.14
MOM 12/9	0.415	0.414	0.999	0.999	0.01
MOM 12/12	0.297	0.294	1.000	1.000	0.00
BM/1	1.272	1.270	1.001	1.001	0.79
BM/3	0.918	0.915	1.001	1.001	0.69
BM/5*	0.741	0.734	1.002	1.002	0.69
CP/1	2.015	2.003	1.009	1.008	0.00
CP/3**	1.240	1.248	1.007	1.007	0.20
CP/5**	1.004	1.016	1.012	1.012	0.55
EP/1	1.464	1.469	0.991	0.997	0.05
EP/3	1.032	1.025	0.998	0.998	0.24
EP/5	0.787	0.785	1.002	1.002	0.43
SALES/1**	0.780	0.778	1.002	1.002	0.48
SALES/3**	0.592	0.588	1.002	1.002	0.19
SALES/5**	0.509	0.504	1.005	1.005	0.28

* Constrained Statistical Arbitrages at the 10% level

** Constrained Statistical Arbitrages at the 5% level

Table 5
Transactions Costs Adjusted, Constrained Mean (CM) Tests of Statistical Arbitrage

Parameter estimates and corresponding p values for constrained test of statistical arbitrage after adjustment for transactions costs. Sample period is from January 1965 to December 2000. Portfolio MOM x / y denotes a momentum portfolio based on returns in the past x months and a holding period of y months. Portfolio CP / y denotes a value portfolio based on past year stock cash flow-to-price and a holding period of y years. Portfolio SALES / y denotes a value portfolio based on past three year sales growth and a holding period of y years. The risk free asset is used to finance the portfolio. H1 and H2 denote the p values from statistical arbitrage tests which test whether the portfolio's mean monthly incremental profit is positive and whether its time-averaged variance is declining over time. The sum of the H1 and H2 columns is the p value for the statistical arbitrage test. Note that the sum may exceed one because of the Bonferonni inequality. Turnover is the number of round trip transactions per month over the number of stocks held in the portfolio. All parameter estimates are calculated from portfolio incremental profit after deducting for estimated transactions costs per month. Estimated transactions cost per month is estimated round-trip transactions cost * turnover where round-trip transactions cost = 1.34%, the transactions costs for the average complexity stock traded in the NASDAQ (Chan and Lakonishok, 1997).

Portfolio	Turnover (%)	μ (mean)	σ (std dev)	λ (growth rate of std dev)	H1 ($\mu > 0$)	H2 ($\lambda < 0$)	Sum (H1+H2)	Statistical Arbitrage
MOM 3/9 [^]	9.26	0.0010	0.025	-0.1082	0.086	0.001	0.088	Yes*
MOM 3/12 [^]	7.22	0.0024	0.023	-0.1316	0.000	0.000	0.000	Yes**
MOM 6/6 [^]	14.20	0.0024	0.037	-0.1252	0.005	0.000	0.005	Yes**
MOM 6/9 [^]	9.49	0.0043	0.026	-0.0908	0.000	0.004	0.004	Yes**
MOM 6/12 [^]	7.23	0.0033	0.030	-0.1732	0.000	0.000	0.000	Yes**
MOM 9/12 [^]	7.24	0.0022	0.022	-0.1254	0.000	0.001	0.001	Yes**
CP/3 [^]	2.17	0.0033	0.132	-0.2273	0.060	0.000	0.060	Yes*
CP/5 [^]	1.46	0.0035	0.131	-0.2685	0.034	0.000	0.034	Yes**
SALES/1 [^]	4.07	0.0080	0.043	-0.0857	0.000	0.019	0.019	Yes**
SALES/3 [^]	2.41	0.0053	0.037	-0.0936	0.000	0.028	0.028	Yes**
SALES/5 [^]	1.52	0.0044	0.042	-0.1517	0.000	0.001	0.001	Yes**

* Significant at the 10% level

** Significant at the 5% level

[^] Point estimates of μ and λ consistent with statistical arbitrage